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Chapter 3

Instruction Organisation in Memory

1

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Chapter 3

Instruction Organisation in Memory

Characteristics of Instructions and the ARM Programmer's Model

Learning Objectives (3.1)

1. Describe the characteristics and role of an instruction.
2. Describe the function of the ARM instruction set **MOV** instruction
3. Describe the ARM programmer's model.
4. Describe the functions and interpretation of several ARM registers.

2

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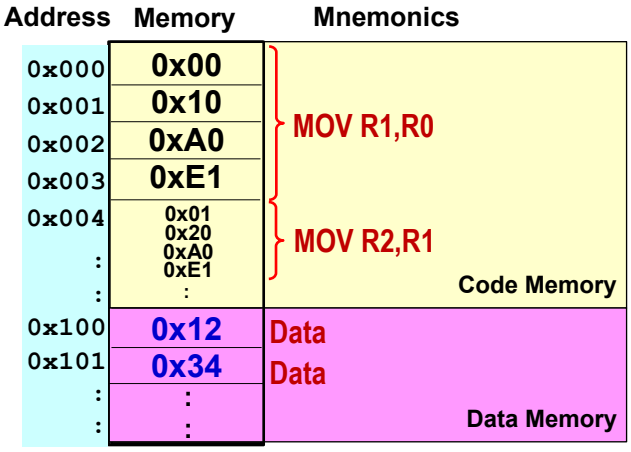
Code and Data in Memory

- Instructions are stored in memory as binary patterns called **machine codes**.
- They are also represented in more readable **mnemonics**.
  - ARM (32-bit CPU) example:
    Note: ARM instructions are **32 bits** long

```
MOV R1,R0
MOV R2,R1
:
```

  - Memory is partitioned into separate **code** and **data** segments.



3

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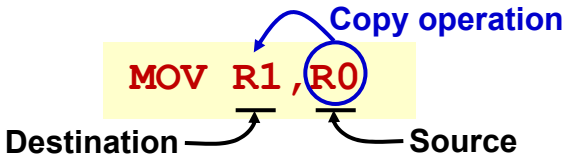
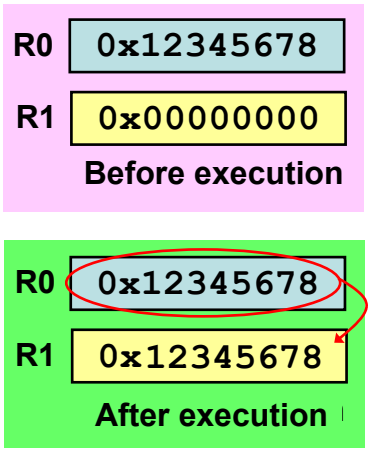
ARM Instruction Format (Introduction)

- Introduction to a simple ARM mnemonic.
  - The `MOV` operator is a two-operand instruction that copies the source operand to the destination operand.
  - The right operand is the source and left operand is the destination.





  - In this example, the both operands are **registers** in the ARM CPU.

4

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Instruction Organization in Memory

- Instructions are binary encoded and stored in memory. Unlike data, instruction bytes tell CPU what actions to take (i.e. they are **executable**).
- Most instruction formats consist of **two** parts.

  - First part of instruction (**op-code**) specifies the operation to be carried out (e.g. move, add, subtract).
  - Remaining bits (**operands**) specify the data itself or the location of data and where results is to be stored.
  - Some operations produce no storable result but may alter program sequence or influence status flags (e.g. **N**, **Z**, **V**, **C**).

5

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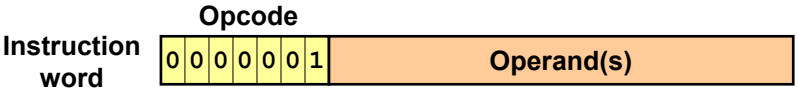
Opcode and Operands

- **How are opcode encoded in an instruction?**



  - If there are 80 different operations, (e.g. `add`, `sub`, etc) then at least 7 bits ($2^6 < 80 < 2^7$) of opcode is needed to represent all the unique bit patterns.
  - The **more variety** of operations supported by the instruction set, the **longer the length** of each instruction.
- **Are there many ways to specify the operand(s)?**
  - Numerous. An operand can be stored as part of the **instruction**, stored in a **register** or stored in **memory**.
  - The method by which the operand is specified is known as **addressing mode**.

6

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Addressing Mode Examples

Addressing Mode	ARM	Intel
Absolute (Direct)	None	`MOV AX,[1000h]`
Register Direct	`MOV R1,R0`	`MOV AX,DX`
Immediate	`MOV R1,#3`	`MOV AX,0003h`
Register Indirect	`LDR R1,[R0]`	`MOV AX,[BX]`
Register Indirect with Offset	`LDR R1,[R0,#4]`	`MOV AX,[BX+4]`
Register Indirect with Index	`LDR R1,[R0,R2]`	`MOV AH,[BX+DI]`
Implied	`BNE LOOP`	`JMP -8`

7

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Role of Instructions

- The role of an instruction is to make purposeful **changes** to the **current state** of the processor.
  - The visible current state of a processor is defined by the **programmer's model** and contents in **memory**.
  - There are three broad categories of instructions for general programming available in a processor:

Data Processing

ARM examples:

```
ADD R0,R1,R2
SUB R1,R2,#3
EOR R3,R3,R2
```

Data Transfer

ARM examples:

```
MOV R1,R0
STR R0,[R2,#4]
LDR R1,[R2]
```

Program Control

ARM examples:

```
B   Back
BNE Loop
BL  Routine
```


8

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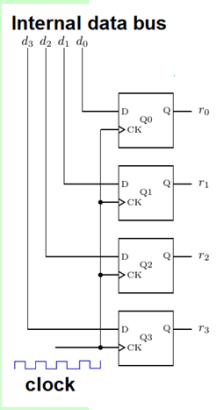
Typical Programmer's Model & Memory

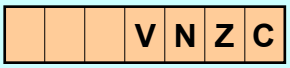


Register circuit

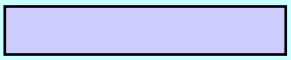
Programmer's model

Registers

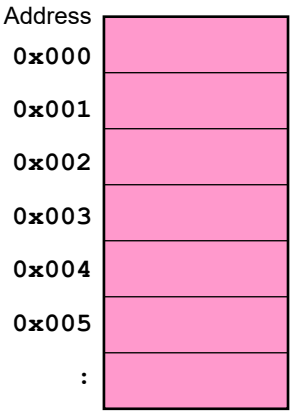
Stack Pointer (SP)



Status Register (SR)



Program Counter (PC)



Memory

9

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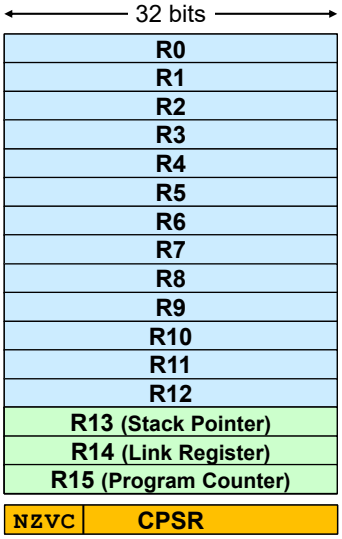
The ARM Programmer's Model

- The ARM processor can operate in various modes. The following are visible registers in the **User mode**.
  - There are 16 32-bit registers.
  - R0 – R12 are general purpose registers.
  - R13 is the Stack Pointer (**SP**).
  - R14 is the Link Register (**LR**).
  - R15 is the Program Counter (**PC**)
  - The Current Processor Status Register (**CPSR**) holds the condition code bits (**NZVC**)



10

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Program Counter

- ARM's register **R15** in the is the Program Counter (**PC**) and it keeps track of **program execution.**
  - The content of the **PC** is an **address**.
  - This address is the start address of the **next instruction** to be fetched.
  - The **PC automatically increments** by the length of the instruction executed (i.e. increment by 4 in the ARM CPU).
  - Sequential execution can be altered by modifying the contents of the **PC** to a new address location (i.e. a jump or a branch operation)
  - Due to **pipeline** architecture of the ARM CPU, the value in the **PC** is the address of the current instruction being executed plus 8 bytes (i.e. 2 instructions).

11

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Stack Pointer and Link Register

- Stack Pointer (**SP**).
  - By convention, **R13** in the ARM processor is designated the stack pointer.
  - The **SP** is used to maintain a space in memory **(stack)** that is used to **temporarily** stored away register information which will be needed again later.
- Link Register (**LR**).
  - **R14** is used as a Link Register during the calling of subroutines.
  - **R14** gets a copy of the **PC** (R15) when a Branch with Link (`BL`) instruction is executed.
  - At all other times, **R14** can also act as a general purpose register.

Learn more: Google Search "ARM stack pointer" or "Link Register"

12

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CPSR Condition Code Flags

CPSR bit(s)	Name	Description NZVC CPSR
31	N	Can set if last operation produced a **negative** result
30	Z	Can set if last operation produced a **zero** result
29	C	Can set if last operation produced a **carry out** in the most significant bit
28	V	Can set if the last operation produced an **overflow** for a **signed** arithmetic operation

N – Negative; **Z** – Zero; **C** – Carry ; **V** – Overflow

13

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Summary

- An instruction usually consist of an **opcode** and an **operand**.
- Instructions in a program change the **states** of a processor as it executes.
- The states of the processor consist of values in:
  - general purpose registers (e.g. **R0 – R12**)
  - specialised function registers (e.g. **CPSR, SP**, **LR** and **PC**)
  - memory contents
- The design of a program is to ensure that these **states changes** in a purposeful manner during the execution of the sequence of instructions.

14

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Chapter 3

Instruction Organisation in Memory

Basic Execution Cycle

Learning Objectives (3.2)

1. Describe the function of a simple **LDR** instruction from the ARM instruction set.
2. Describe the basic execution cycle of a simple **LDR** instruction.
3. Describe the factor that determines the execution speed of an instruction.

15

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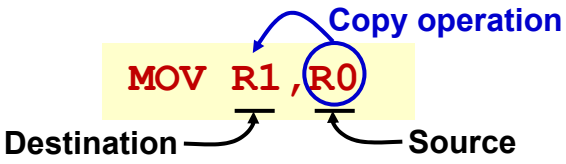
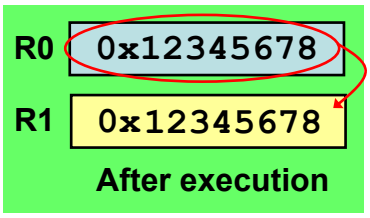
The MOV Instruction (Review)

- The `MOV` operator copies **register** content to a register.





16

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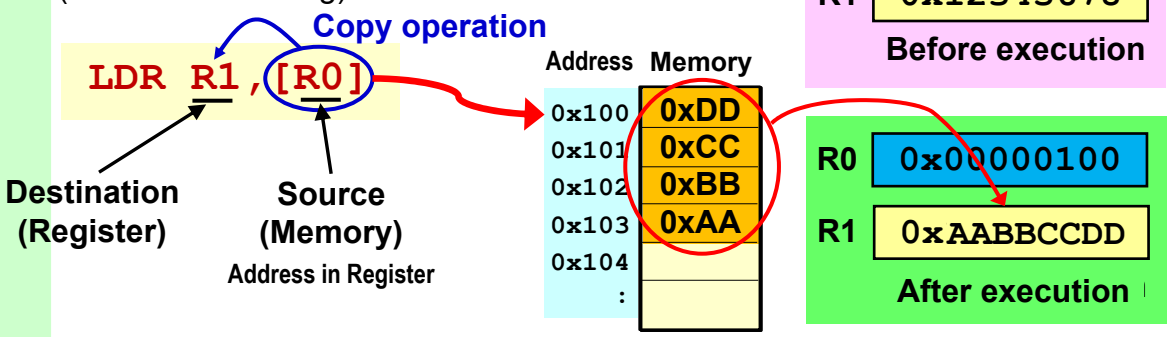
The LDR Instruction

- The `LDR` operator copies **memory** content to a register.
  - The left operand is always a destination register.
  - The right source operand is a memory location whose address is contained in a register (indirect addressing).



17

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The Role of the Processor

- What is the role of the processor (CPU)?
  - **Fetch instructions** - CPU reads instructions from memory.
  - **Decode instructions** - Instruction is decoded to determine action required.
  - **Fetch data** – Some data from memory may need to be fetched in order to complete execution of instruction (optional).
  - **Execute** – Instruction execution may require CPU to perform some arithmetic or logical operation on the data, or just move the data into a register.
  - This **fetch-decode-execute** cycle is performed repeatedly by the CPU once powered-on.

18

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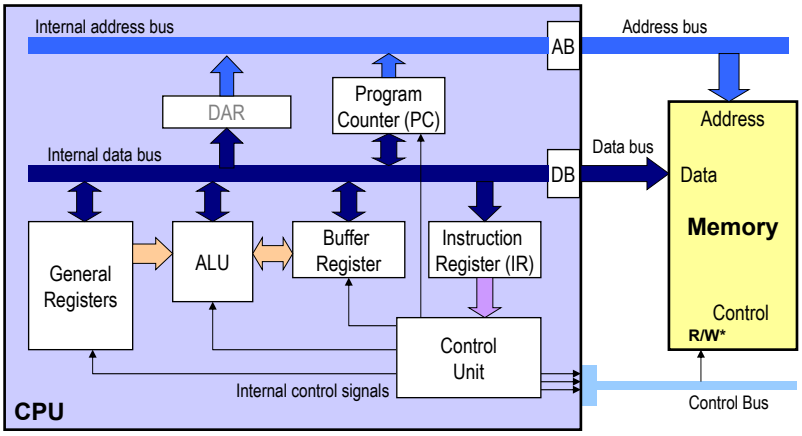
Basic Execution Cycle

A Simple Processor

- Basic components of a simple processor (CPU) and its interface signals to external main memory.



19

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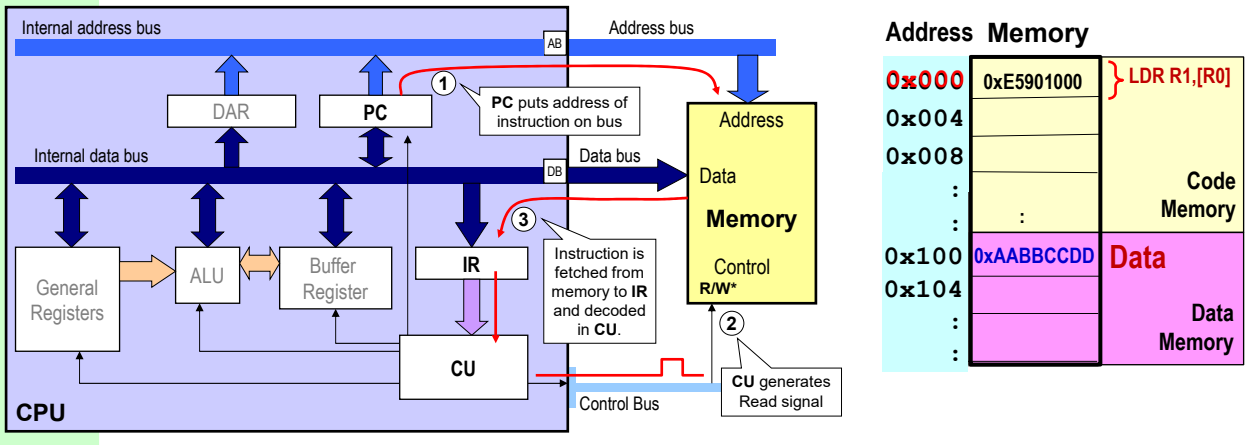
Basic Execution Cycle

Fetch Cycle - Instruction

- Consider an instruction like: `LDR R1,[R0]`.
  In the fetch cycle, the **PC** points to address of next the instruction and its opcode is fetched into the **IR**.



20

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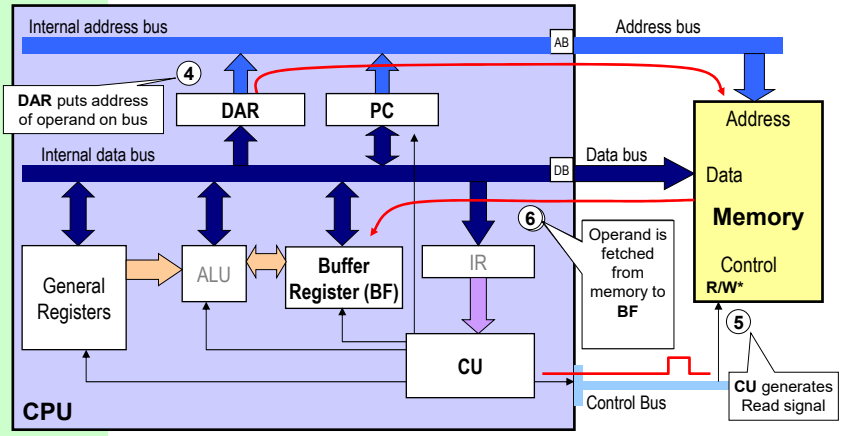
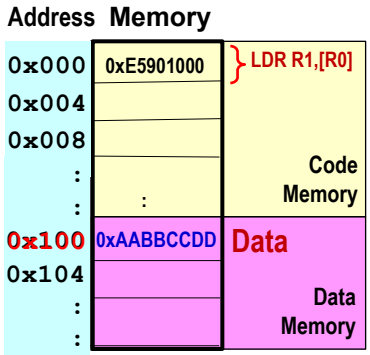
Basic Execution Cycle

Fetch Cycle - Operand

- Decoding `LDR R1,[R0]` suggest an operand is needed. Since its is stored in memory, another fetch is required to retrieve operand into CPU.





21

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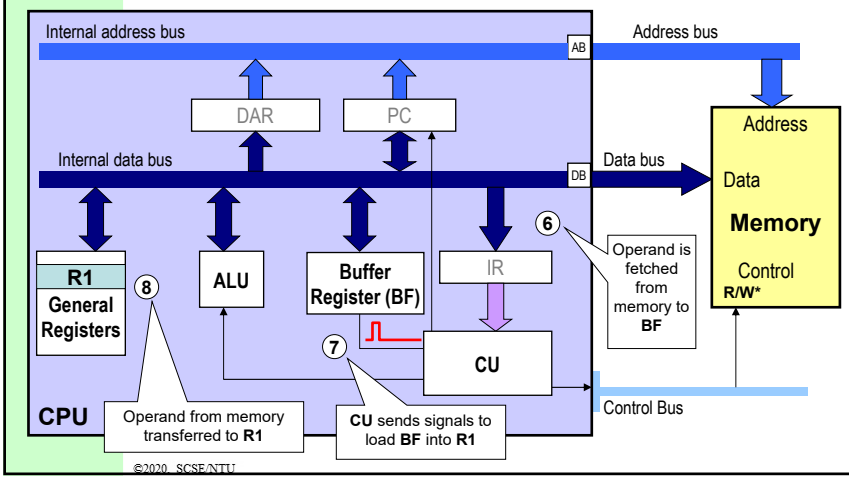
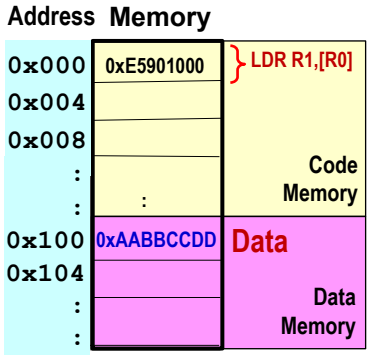
Basic Execution Cycle

Execute Cycle - Load

- In the execution cycle of `LDR R1,[R0]`, the operand fetch from memory is loaded into the destination register **R1**.





22

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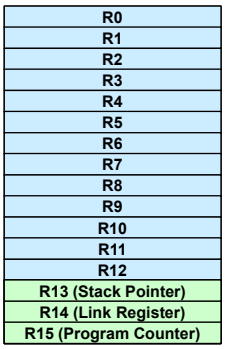
Program Execution (Summary)

- Execution of a single instruction may involve multiple accesses to memory. In most single memory (von Neumann-type) CPU, different instructions take **different number** of clock **cycles** to execute.
- Data transfer on external bus is slower than within CPU's internal bus. μP system performance is limited by data traffic bandwidth between CPU and memory (also known as the **von Neumann bottleneck**).
- Keeping regularly used operands in CPU **registers** helps reduce memory access.
- Keeping instructions and data in separate memories (**Harvard architecture**) can help make instructions execute in more regular cycles (using parallel fetches) and thus improve performance.



23